STATISTICS

Chapter 7 Homework

Sampling distributions and the CLT

Sections 7.1-7.3 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Individuals, averages, and sums

1. A population of completion times has mean 12 minutes and SD 6 minutes and is moderately right-skewed without extreme outliers. Random observations are independent. For samples of 36, find the mean and standard error of the sample mean. Explain why a normal approximation is reasonable.

2. Distinguish the population SD from the standard error in Problem 1. Which describes spread among individual times, and which among repeated sample means?

3. For samples of 36, approximate P(sample mean > 13). Sketch and label the sampling distribution. Use $P(Z \leq 1) = 0.8413$.

4. For samples of 36, approximate P(11.5 < sample mean < 12.5). Use $P(Z \leq 0.5) = 0.6915$ and $P(Z \leq -0.5) = 0.3085$.

5. Find the mean and SD of the sum of 36 times. Rewrite the event “sum exceeds 468 minutes” as an event about the sample mean, and find its approximate probability.

6. Increase the sample size to 144. Find the new standard error. Explain why multiplying sample size by 4 does not divide the standard error by 4.

When approximations apply

7. Can the normal approximation for the sample mean in Problem 1 be used to calculate P(one individual time > 13)? Explain. What additional information would be needed?

8. A team takes 36 successive readings that are strongly correlated. Explain why inserting $n = 36$ in the usual standard-error formula may underestimate uncertainty.

9. For samples of 36 in Problem 1, find the 95th percentile of the sample mean using $z = 1.6449$. Explain what repeated samples have to do with this percentile.

10. Two binomial settings have the same large number of trials. In the first, successes and failures are both common. In the second, successes are so rare that only a few are expected. Which setting better supports a normal approximation? Explain why a large trial count by itself is insufficient, and why the normal model represents a single count with a small interval.

11. A student says that increasing sample size makes the original population bell-shaped and removes selection bias. Correct both claims.